

ABHINAV TUPE

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EDUCATION

MIT World Peace University

Bachelor of Technology in Computer Science (Business System) | CGPA:8.19

Pune, Maharashtra

Aug 2023 - May 2027

Academic Background

Class X: 91% (ICSE), Class XII: 93% (CBSE)

ABOUT ME

Computer Science undergraduate with hands-on experience in AI/ML, backend development, and data analytics. Skilled in building scalable AI-powered systems using Python, FastAPI, SQL, and machine learning frameworks for anomaly detection, predictive analytics, and realtime monitoring. Experienced in developing REST APIs, data processing pipelines, and analytics-driven applications using structured and unstructured datasets. Strong interest in financial analytics, AI-driven decision systems, and enterprise-scale data science solutions.

PROJECTS

FINSIGHT AI - Financial Risk & Compliance Intelligence Platform

- Tech Stack: Python, FastAPI, SQLite, Scikit-learn, Streamlit, Pandas
- Built a financial analytics platform for fraud detection, anomaly monitoring, and transaction risk analysis using structured banking datasets
- Developed anomaly detection and risk-scoring pipelines using Isolation Forest and Random Forest models with feature engineering on transaction data
- Designed interactive analytics dashboards for suspicious transaction tracking, KPI visualization, fraud insights, and operational monitoring
- Implemented compliance analysis workflows using keyword-based NLP techniques and FastAPI REST APIs for transaction risk prediction and reporting
- Built ETL pipelines for transaction preprocessing, data storage, and analytics workflows using Pandas and SQLite

AI-Powered Cloud Monitoring & Anomaly Intelligence Platform

- Tech Stack: Python, Flask, AWS EC2, Docker, GitHub Actions, SQLite, REST APIs
- Built a production-style cloud monitoring and anomaly intelligence system for realtime log ingestion, operational analytics, and automated alert generation
- Developed REST APIs for monitoring workflows, dashboard analytics, health-check services, and alert management using Flask
- Implemented anomaly detection pipelines for service failures, timeout analysis, repeated error tracking, and high-frequency event monitoring
- Designed Dockerized CI/CD deployment workflows with GitHub Actions and AWS EC2 integration for scalable cloud deployment automation
- Integrated automated alert systems, persistent event logging, and realtime operational monitoring pipelines

VisionFlowAI – Realtime AI Analytics Platform

- Tech Stack: Python, FastAPI, YOLOv8, OpenCV, Redis, PostgreSQL
- Built a realtime AI-powered analytics platform for live event detection and intelligent video processing using YOLOv8 and OpenCV
- Designed asynchronous event-processing pipelines using Redis queues for scalable realtime analytics workflows
- Developed FastAPI REST APIs for event ingestion, detection management, analytics reporting, and system monitoring
- Implemented PostgreSQL-backed storage architecture for scalable event tracking and analytics processing
- Optimized low-latency realtime inference pipelines for continuous AI-driven event monitoring

RELEVANT COURSES

Data Structures C Algorithms, Operating Systems, DBMS, Computer Networks, OOP, Machine Learning, Data Mining, Cloud Microservices Applications, Modern Web Development

TECHNICAL SKILLS

- Programming: Python, C++
 - Cloud & DevOps: AWS (EC2, S3, IAM), Terraform, CI/CD, GitHub Actions, Docker
 - Backend Development: Flask, REST APIs
 - Infrastructure & Automation: Infrastructure as Code (IaC), Cloud Deployment, Environment Provisioning
 - Databases: MySQL, SQLite
 - Tools & Platforms: Git, GitHub, Linux Basics
 - Monitoring & Governance: CloudWatch (basic), Logging, Cloud Security Fundamentals
- Concepts: Cloud Computing, DevOps, Networking Basics, Automation Workflows

ACHIEVEMENTS

Software Engineer Intern — Rikouu Healthcare Private Limited
[Jan 2026 - Mar 2026]